Supplementary material

Apparent sequence-model contribution depends strongly on negative-set construction: a calibration across 94 ENCODE eCLIP datasets

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How to read this document

Ten supplementary figures and one supplementary table. Each figure carries a legend giving the panels, the sample size, the uncertainty definition and the committed table the values are read from, so that no figure needs the main text to be interpretable.

Two conventions hold throughout and are not repeated in every legend.

The estimand. The *nested contribution* of a model under a protocol is the pooled out-of-fold AUROC of a logistic model on 19 composition features plus that model's score, minus the AUROC of the 19 features alone. Composition is mononucleotide and dinucleotide frequencies plus Shannon entropy, with one level dropped from each frequency family because frequencies sum to one and keeping every level makes the design matrix singular: $3 + 15 + 1 = 19$ columns. Higher means the sequence model adds more over knowing base composition.

Uncertainty. Unless a legend says otherwise, intervals are 95% percentile intervals from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. The panel is 94 datasets over 79 proteins, 15 of which appear in both cell lines, so resampling datasets would treat correlated rows as independent evidence.

Estimator. Figures showing a span or a contribution use the two-stage estimator unless stated, which is what this literature computes. The paper's primary estimator is the cross-fitted one; the two differ by about a ninth of the span and the main text reports both.

Figure and table numbering

The repository stores figures under their build names. This mapping is generated and checked by `tests/unit/test_supplement.py`, so it cannot drift from the files.

supplementary	repository file (in `results/figures/`)	subject	section
Figure S1	`f0_panel_overview`	the panel and how it was drawn	Methods
Figure S2	`f1_cost_of_matching`	the two-protocol contrast	3.1
Figure S3	`f2_four_models`	four model classes, one protocol	Methods
Figure S4	`f3_strand_placebo`	the strand control	Limitations
Figure S5	`f4_replication`	cross-cell-line replication	3.1
Figure S6	`f5_size_modification`	effect modification by size	3.1
Figure S7	`f6_k_sweep`	sensitivity to k	Methods
Figure S8	`f7_region`	region heterogeneity	3.1
Figure S9	`f8_scale_check`	the compression correction	3.2
Figure S10	`f13_baseline_order_models`	order-3 baseline by model class	3.3
Table S1	`supplementary_table_s1.csv`	dataset to ENCODE accession	Data avail.

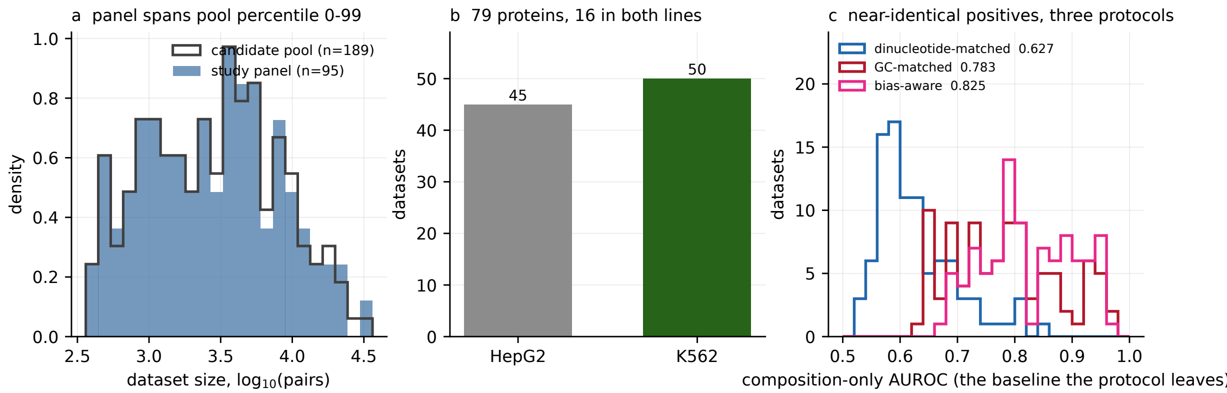

Figure S1: The study panel, and the check that it is not size-biased. (a) Size distribution of the 94 study datasets, as $\log_{10}$ matched pairs, against the candidate pool they were drawn from, shown as a step outline. The panel was selected as a deterministic systematic subset by pair rank, which guarantees coverage of the size range; it is not a probability sample and cannot be assumed free of aliasing against structure ordered by pair count. **(b)** Composition of the panel by protein and cell line. 79 distinct proteins, of which 15 are assayed in both K562 and HepG2 and therefore contribute two datasets each. This is why every interval in the study resamples proteins rather than datasets. **(c)** Composition-only AUROC under each of the three protocols, one point per dataset. This is the independent variable of the study: the baseline each protocol leaves for a sequence model to improve on. **n**: 94 datasets, 79 proteins, 2 cell lines. **Uncertainty**: none plotted; panels show distributions rather than estimates. **Source**: `panel_summary.csv`, `candidate_sizes.csv`, `three_arm_per_dataset.csv`.

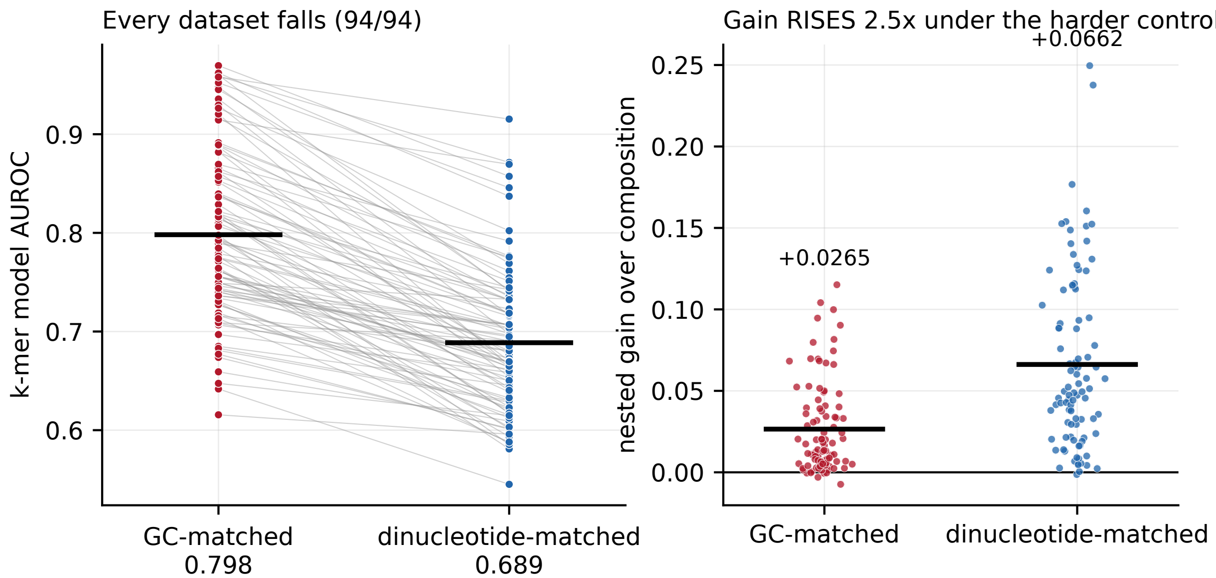

Figure S2: Moving from GC-matched to dinucleotide-matched negatives lowers apparent AUROC and raises measured contribution. Paired within dataset, because the two arms share nearly all of their positives; an unpaired plot would discard the structure that makes the per-dataset counts meaningful. **(a)** The 4-mer model's own AUROC under each protocol. It falls in every dataset, which is expected: dinucleotide matching removes the compositional cue the model was partly using. **(b)** The nested contribution over composition under each protocol. It rises. **What this figure is not.** The two arms do not have identical positives and the model is not held fixed across them: it is refitted under each protocol, and each matcher rejects the positives it cannot pair, leaving a median Jaccard of 0.9972 between the two positive sets and exact agreement in only 10 of 94 datasets. **n**: 94 paired datasets. **Uncertainty**: not plotted; the panel-level intervals are in the main text. **Source**: `cost_of_matching.csv`.

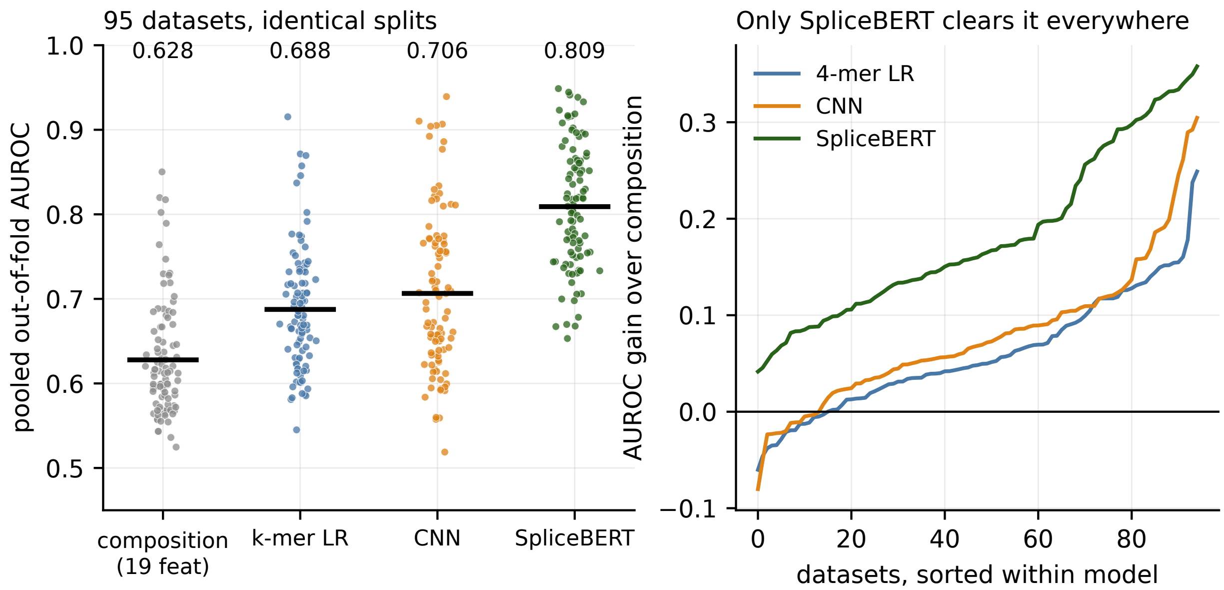

Figure S3: Four model classes under one protocol, on identical rows and folds. **(a)** Pooled out-of-fold AUROC for the 19-feature composition baseline, the 4-mer logistic model, the convolutional network and fine-tuned SpliceBERT. Within an arm all four are scored on the same rows under the same chromosome-grouped partition. **(b)** The same four as a gain over the composition baseline, with datasets sorted within each model. Only SpliceBERT exceeds the baseline in every dataset. **Why it is supplementary.** This is a methods orientation figure rather than a result: it establishes that the model ladder behaves as expected under one protocol, which is the premise the three-protocol comparison in the main text varies. **n**: 94 datasets. **Uncertainty**: not plotted; each point is one dataset's pooled out-of-fold AUROC. **Source**: `matched_four_models.csv`.

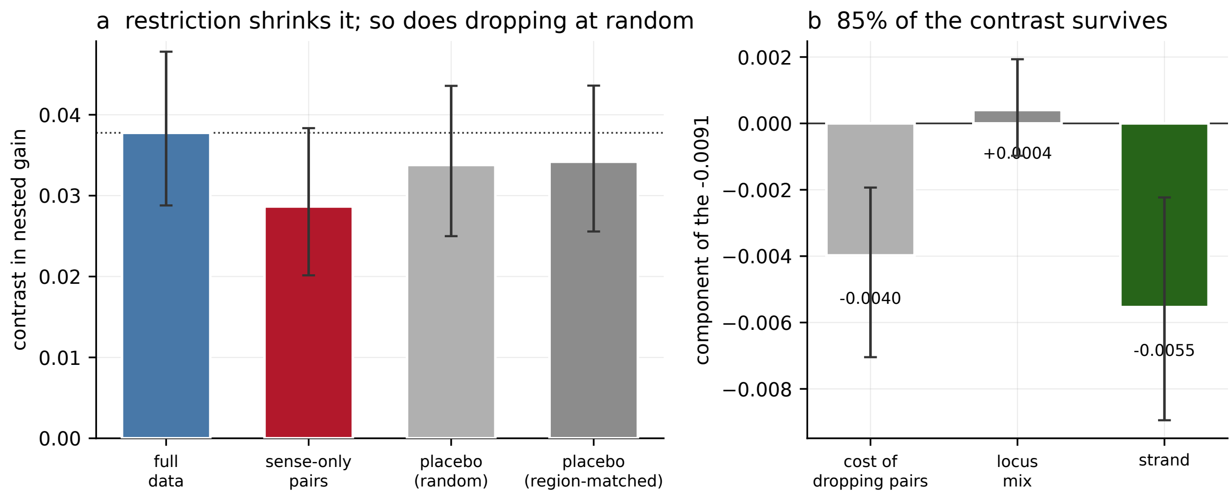

Figure S4: The strand control, and why the naive version of it overstates the artefact. Negatives inherit the strand of the positive they are matched to rather than the strand of the gene they sit in, so a fraction of them are antisense. Restricting to pairs whose negative is unambiguously on its own gene's strand discards most of the data, and a 256-feature model loses more from that restriction than a 19-feature baseline does, in both arms. **(a)** The contrast under four conditions: the full data, the sense-only restriction, a placebo dropping the same number of pairs at random, and a placebo matched on transcript region crossed with gene-density quartile. The figure is the argument: the naive control attributes the whole shrinkage to strand, and the two placebos show why it should not. **(b)** The same information as a decomposition of the -0.0091 that restriction alone reports, into the cost of dropping pairs at all and the strand-specific remainder. **Why gene density.** Retention requires exactly one overlapping gene strand, so it selects against multi-gene loci by construction, and the region-only placebo does not remove that. **Result.** The strand-specific excess is -0.0055 and 85.3% of the contrast survives correction. **n**: 40 datasets, over 40 distinct proteins, so protein clustering is a no-op here and is asserted rather than assumed. 20 placebo seeds per dataset per arm. **Uncertainty**: 95% percentile intervals over datasets. **Source**: `strand_placebo.csv`.

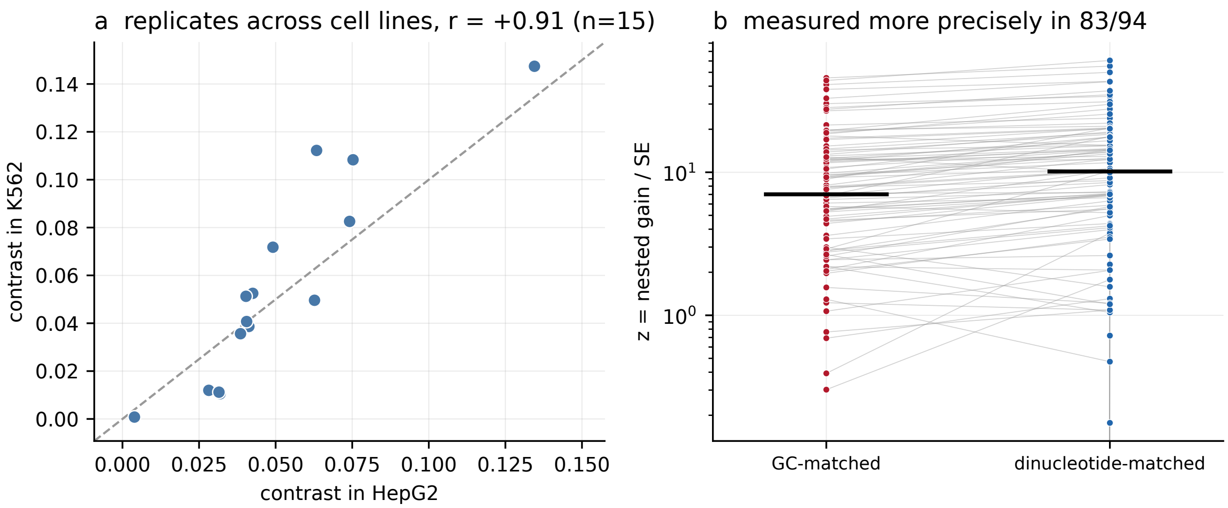

Figure S5: The contrast replicates across cell lines, and is measured more precisely under the harder protocol. (a) For the 15 proteins assayed in both K562 and HepG2, the contrast measured in one cell line against the contrast measured in the other. These are separate eCLIP experiments with separately drawn negatives, so one line is an out-of-sample prediction of the other. Pearson $r = +0.9094$. The design forces the sign in each cell line independently and guarantees nothing about whether the magnitudes agree, which is what makes this evidence rather than arithmetic. **(b)** $z = \text{gain/SE}$ under each protocol, paired per dataset. The gain rises faster than its standard error, so the same contribution reaches the same confidence with fewer labelled windows. **A caveat that belongs with panel (a).** Partialling out each dataset's total nested gain reduces the correlation to $+0.3321$ with $p = 0.2265$, so the replication is not independent of overall signal strength. **n**: 15 proteins for (a), 94 datasets for (b). **Uncertainty**: Pearson correlation with a protein-level bootstrap interval in the main text; DeLong standard errors for z. **Source**: `r1_robustness.csv`, `cost_of_matching.csv`.

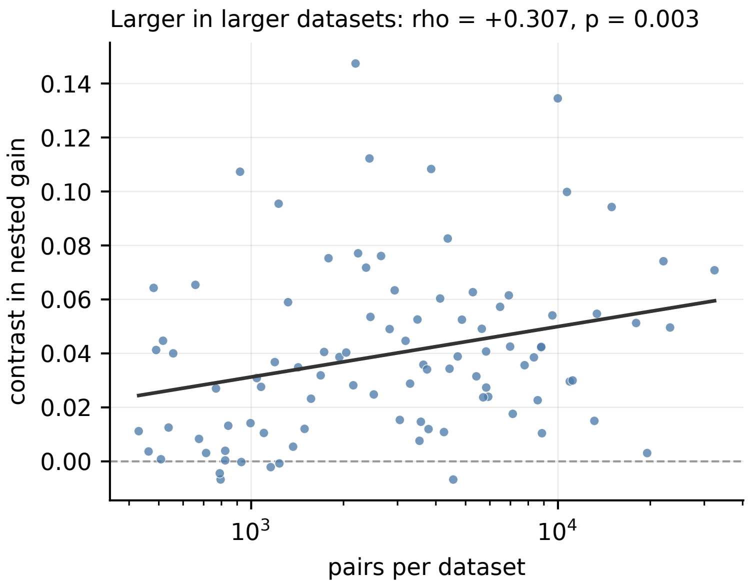

Figure S6: The contrast is larger in larger datasets. Each point is one dataset: matched pairs on a logarithmic axis against the contrast in nested contribution between the dinucleotide-matched and GC-matched protocols. **Why this cannot be a confound.** The contrast is paired within dataset, so both arms have identical n and size affects precision rather than bias. It is reported here as effect modification, which is a different and weaker statement: the protocol matters more where there is more data. **n**: 94 datasets. **Uncertainty**: Spearman correlation, printed in the panel title; the protein-clustered interval is in the main text. **Source**: `cost_of_matching.csv`, `robustness.csv`.

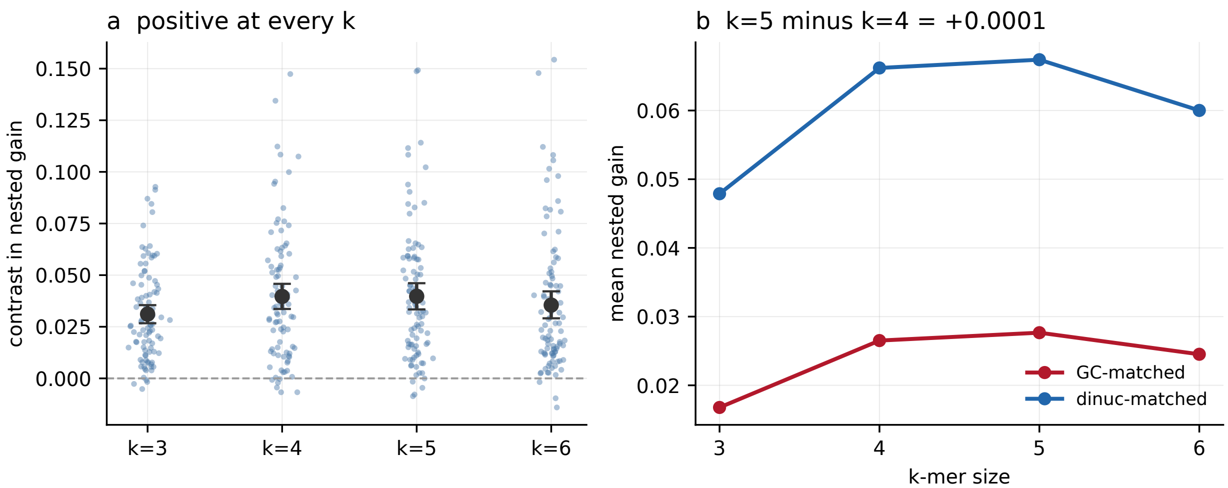

Figure S7: The contrast does not depend on the choice of k. The composition baseline and the k-mer model are refitted from raw sequence at four orders for all 94 datasets in both arms, rather than read from the tables the main analysis wrote. Together with `recompute.py` this is one of only two results in the study that are re-derived along a different path rather than compared against a record of themselves; the rebuild reproduces the published contrast to 1.2×10^{-6}. **(a)** The contrast at $k = 3, 4, 5, 6$, one point per dataset. Positive at every order, and positive at every order in 82 of 94 datasets individually. **(b)** The mean nested gain in each arm as a function of k, so a reader can see where the contrast comes from rather than only its difference. **What it settles.** The manuscript described the model as a 5-mer in four places before a referee reproduced the tables at $k = 4$ and found the error. The difference between the two is +0.0001, paired Wilcoxon $p = 0.84$. The contrast is also smallest at $k = 3$, so order 4 is not the most favourable choice available. **n**: 94 datasets per order, both arms. **Uncertainty**: 95% protein-clustered percentile intervals in the source table. **Source**: `k_sweep.csv`, `k_sweep_per_dataset.csv`.

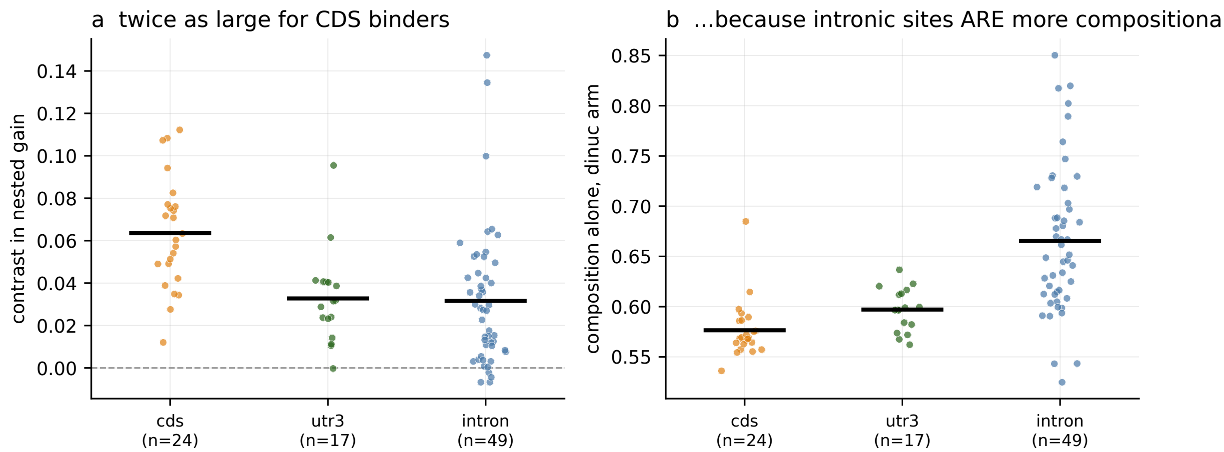

Figure S8: The contrast is larger for datasets whose binding sites are mostly coding, and the mechanism is checkable in the same figure. Datasets are grouped by the transcript region their positives mostly occupy. **(a)** The contrast by dominant region. CDS-dominant datasets give +0.0635 against +0.0316 for intron-dominant ones, a difference of +0.0319, which is as large as the headline contrast itself. **(b)** The proposed mechanism, plotted rather than asserted: intronic sites are compositionally distinctive, so composition alone already discriminates them and leaves the protocol less to expose. Composition-only AUROC is 0.6656 for intron-dominant datasets against 0.5765 for CDS-dominant ones. If this ordering ever inverted, the explanation in the text would be wrong, and a gate asserts it. **The limitation, which is gated too.** The intronic fraction correlates with the contrast at −0.3399, but partialling out total nested gain removes the association. Region indexes how much non-compositional signal exists; it is not an independent mechanism. **n**: 94 datasets, grouped into three region classes with at least 8 datasets each. **Uncertainty**: Mann–Whitney and Kruskal–Wallis tests in the source table. **Source**: `region_heterogeneity.csv`, `region_heterogeneity_per_dataset.csv`.

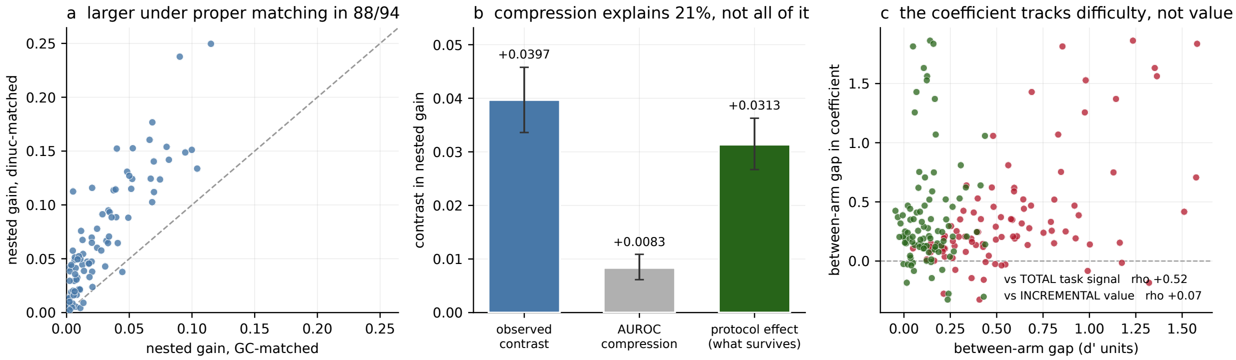

Figure S9: How much of the contrast is compression against a higher baseline, and why the decomposition is reported as a range rather than a number. An AUROC increment is bounded above by the headroom the baseline leaves, so part of any comparison between arms with unequal baselines is arithmetic. The transplant converts AUROC to d', moves one arm's increment onto the other arm's baseline, and asks what remains. **(a)** The contrast per dataset, which needs no transplant and no link function: it is a difference of two AUROC differences on the same rows. **(b)** The transplant. Compression is real, a factor of 1.5081, and does not account for the contrast: the protocol effect net of scale is +0.0313 in one direction. **(c)** The diagnostic behind the log-odds reversal. The between-arm coefficient gap tracks each task's total signal rather than the incremental value it should measure, which is the standard non-comparability of logistic coefficients across separate fits. **Why the paper declines to quote a single number.** Running the transplant in the other direction gives +0.0215, and across six monotone links and both directions the smallest member is +0.0188. Every member keeps the sign, so the contrast is not a compression artefact; the width is why the decomposition is a sensitivity analysis rather than a headline. Separately, the identifying assumption that the d' increment is baseline-invariant is refuted by its own diagnostic, so no model's protocol effect is point-identified. **n**: 94 datasets. **Uncertainty**: 95% protein-clustered percentile intervals. **Source**: `scale_check.csv`, `protocol_identification.csv`.

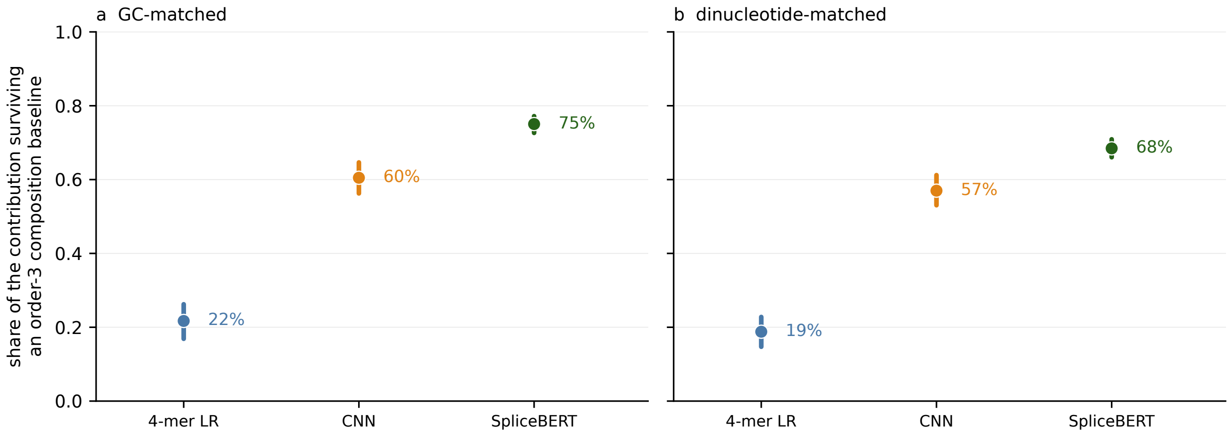

Figure S10: Raising the composition baseline from order 2 to order 3, for all three model classes. The composition baseline stops at dinucleotides while the headline model is a 4-mer, so the measured quantity is short-range compositional signal one order above wherever the matcher stopped. Where the baseline stops is an analyst's choice, and this figure measures what that choice costs. The extended baseline adds all 64 trinucleotide frequencies, 82 features in total. **(a)** GC-matched arm and **(b)** dinucleotide-matched arm: the share of each model's contribution that survives an order-3 baseline. For the 4-mer the surviving share on the GC arm is 0.2170, against 0.6049 for the convolutional network and 0.7504 for SpliceBERT. **The shares are not the result, and reading them alone is a mistake this analysis made once.** The *absolute* amount absorbed is statistically indistinguishable across model classes, and the convolutional network loses the least of the three, which no capacity story predicts. The shares follow from the totals. What the figure supports is that the protocol contrast survives an order-3 baseline for every model class, not that the 4-mer is uniquely fragile. **n**: 94 datasets, three model classes, two arms. **Uncertainty**: 95% protein-clustered percentile intervals in the source table. **Source**: `baseline_order_models.csv`, `baseline_order_models_per_dataset.csv`.

Table S1. Dataset to ENCODE accession

The file `supplementary_table_s1.csv` ships with this document and is committed under `results/tables/`. One row per dataset in the 95-dataset candidate panel, giving the protein, the cell line, the ENCODE file and experiment accessions, the number of replicates, the matched pair count, and a flag for whether the dataset carries all three protocols.

One row has that flag false: NCBP2 in K562 cleared the minimum-pair floor under dinucleotide matching and not under GC matching, so it is in the candidate panel and has no three-protocol result. That is the single blank row in the table and it is the difference between the 95 selected datasets and the 94 the study reports on.